

A mini project report
on
Smart Power Nexus using embedded system
Submitted in partial fulfilment of the requirement for the award of degree
of
BACHELOR OF TECHNOLOGY
in
ELECTRONICS AND COMMUNICATION ENGINEERING

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CERTIFICATE

This is to certify that the main project work entitled “**Smart Power Nexus using embedded system**” is a Bonafide work done by **Malleboyina Harsha Vardhan (23FE1A0486)** , **Pedapudi Yaswanth (23FE1A04B8)**, **Penumala Akhil (23FE1A04B9)**, **Marupudi Prem Kumar (23FE1A0491)**, **Mamidi Neelraj (23FE1A0487)**, under my guidance and submitted in partial fulfilment of requirement for award of degree of **Bachelor of Technology in Electronics and Communication Engineering** by Jawaharlal Nehru Technological University, Kakinada.

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DECLARATION

We hereby declare that the work described in this project work, entitled “**Smart Power Nexus using embedded system**” which is submitted by us in partial fulfilment for the award of **Bachelor of Technology (B.Tech)** in the Department of **Electronics & Communication Engineering** to the Vignan’s Lara Institute of Technology and science affiliated to Jawaharlal Nehru Technological University Kakinada, Andhra Pradesh, is the result of work done by us under the guidance of **Ms. P.V. N. Lakshmi**, Assistant Professor, in the department ECE.

The work is original and has not been submitted for any Degree/Diploma of this or any other university.

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ABSTRACT

1. This project presents Smart Power Nexus, an embedded-based intelligent power distribution and control system designed to provide regulated electrical supply to multiple devices such as mobile phones, electric vehicles, and household electrical appliances. The system ensures safe and efficient power delivery through automated control mechanisms.
2. The system uses a microcontroller-based embedded platform to manage and control the power switching process. Based on predefined conditions and programmed logic, the controller activates or deactivates the power supply to connected loads, ensuring controlled energy access.
3. Smart Power Nexus functions as a centralized energy access unit suitable for public utility stations, campuses, commercial areas, and charging hubs. It supports multi-device compatibility and structured power management.
4. A relay-based switching module provides electrical isolation between the low-voltage control circuit and the high-voltage AC supply, ensuring operational safety and protection against overload conditions. The system minimizes energy misuse and enhances reliability.
5. The project demonstrates the practical application of embedded systems in smart energy automation and infrastructure development. Future enhancements may include IoT-based remote monitoring, digital payment integration, current sensing modules, and real-time energy analytics.

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NOMENCLATURE

Symbol / Abbreviation	Description
Embedded System	A microcontroller-based system designed to control and manage hardware components for specific tasks automatically.
NodeMCU (ESP8266)	A WiFi-enabled microcontroller board used as the main controller of the system to manage power distribution and control operations.
Relay Module	An electromechanical switching device that allows a low-voltage microcontroller to control high-voltage AC power safely.
Power Distribution Unit	A central unit that distributes electrical power to different connected devices such as mobiles, vehicles, and appliances.
AC Power Supply	Alternating current electrical source used to power devices connected to the system..
Charging Port	Output interface where users connect their devices such as mobile chargers or other electrical appliances.
Control Signal	Electrical signal generated by the microcontroller to activate or deactivate switching modules like relays.
Protection Circuit	Safety mechanism such as fuse or MCB used to prevent overcurrent and protect the system from electrical faults.
IoT (Internet of Things)	Technology that enables devices to communicate and exchange data over the internet for remote monitoring and control.
Load	Any electrical device connected to the system that consumes power, such as a mobile charger, electric vehicle charger, or appliance.
Power Management	The process of controlling and optimizing electrical power distribution to ensure efficiency and safety.

CHAPTER 1

INTRODUCTION

1.1 Overview

The **Smart Power Nexus** is an embedded-based intelligent power distribution and control system designed to provide controlled electrical supply to multiple devices such as mobile phones, electric vehicles, and household electrical appliances. With the increasing demand for electricity and the growing number of electronic devices used in daily life, efficient power management has become an important requirement.

In many public places such as railway stations, bus stands, college campuses, shopping complexes, and commercial buildings, users often require reliable access to electrical power for charging and operating their devices. However, uncontrolled power access may lead to electricity misuse, overload conditions, and safety risks.

The proposed Smart Power Nexus system provides a centralized solution for controlled power distribution using an embedded microcontroller platform. The embedded controller manages the switching mechanism and regulates the power supply to connected loads through relay modules. The system ensures that electrical power is delivered safely and efficiently.

This project demonstrates how embedded systems can be applied in power management and automation to create smart infrastructure solutions. The system is simple, cost-effective, and suitable for deployment in public and commercial environments.

1.2 Problem Statement

In modern environments, electrical devices such as smartphones, portable electronics, and electric vehicles are used extensively. Users often face difficulties in accessing reliable power sources in public areas or shared environments. In many cases, available power outlets are either limited, uncontrolled, or unsafe.

Unregulated power access can lead to problems such as energy wastage, electrical overload, and potential safety hazards. Additionally, traditional power outlets do not provide any mechanism for monitoring or controlling power distribution among multiple users.

Therefore, there is a need for a smart and automated power management system that can provide controlled access to electricity while ensuring safety and efficient energy utilization. The Smart Power Nexus project addresses this issue by designing an embedded-based power distribution system that regulates electrical supply and manages multiple connected devices effectively.

1.3 Objectives of the Project

The main objectives of this project are as follows:

1. To design and develop an embedded-based smart power distribution system.
2. To provide controlled electrical power supply to multiple devices such as mobile phones, vehicles, and appliances.
3. To implement safe switching of AC power using relay modules controlled by a microcontroller.
4. To demonstrate the application of embedded systems in smart energy management.
5. To develop a cost-effective and reliable power control solution suitable for public and commercial environments.

1.4 Scope of the Project

The Smart Power Nexus system focuses on providing a centralized power distribution solution using embedded technology. The system controls electrical power delivery to connected loads through a microcontroller and relay-based switching mechanism.

This project can be implemented in locations where controlled access to electrical power is required, such as educational institutions, charging stations, public utility areas, and commercial facilities.

The system can also be expanded in the future by integrating advanced technologies such as IoT-based monitoring, digital payment systems, remote control applications, and energy consumption tracking. These improvements can transform the system into a fully automated smart energy management platform.

1.5 Motivation

The rapid growth of electronic devices and electric mobility has increased the demand for reliable and accessible power sources. In many situations, users struggle to find safe and controlled charging facilities for their devices.

This motivated the development of the Smart Power Nexus system, which aims to provide an intelligent and organized power distribution solution. By integrating embedded technology with power control mechanisms, the system improves energy accessibility while maintaining safety and efficiency.

The project also demonstrates the importance of embedded systems in modern infrastructure and smart energy solutions.

Many accidents have occurred because gas leaks went undetected during the night or when people were away from home. Traditional methods of detection—such as relying on smell—are slow and unreliable, especially for elderly people or in places with poor ventilation. These incidents highlight the urgent need for an automatic, real-time detection and response system that can prevent such disasters.

1.6 Methodology

The development of the Smart Power Nexus system follows a systematic approach involving hardware design, software programming, and system testing.

System Design

The system consists of a NodeMCU (ESP8266) microcontroller, relay modules for power switching, power supply circuits, and output sockets for connecting electrical devices. The microcontroller acts as the central control unit that manages the operation of the entire system.

Working Principle

The embedded controller monitors the system conditions and controls the relay module based on programmed logic. When the system is activated, the microcontroller sends control signals to the relay module, allowing AC power to pass through the output socket.

The relay acts as a switching interface between the low-voltage microcontroller circuit and the high-voltage AC power supply. This ensures safe and reliable operation while delivering electrical power to connected loads.

1.7 Applications

The **Smart Power Nexus** system has a wide range of practical applications in environments where controlled and efficient power distribution is required. Some of the major applications are:

- **Public Charging Stations:**

Used in railway stations, bus stations, airports, and metro stations to provide controlled charging facilities for mobile phones and electronic devices.

- **Educational Institutions:**

Installed in colleges, universities, and libraries where students require charging points for laptops and mobile devices.

- **Commercial Buildings:**

Useful in offices, malls, and shopping complexes to manage shared power outlets and prevent electricity misuse.

- **Electric Vehicle Charging Points:**

Can be used in small-scale EV charging stations for controlling and managing power supply to electric vehicles.

- **Smart City Infrastructure:**

Can be integrated with smart city systems to provide intelligent power management and automated energy control.

- **Hotels and Restaurants:**

Provides convenient charging facilities for customers while ensuring controlled power usage.

- **Hospitals and Healthcare Centers:**

Can be used to provide safe charging access for medical devices and mobile equipment.

- **Industrial Workspaces:**

Useful in workshops and small industries where multiple electrical tools require regulated power supply.

- **Home Automation Systems:**

Can be integrated with smart home systems for intelligent control of electrical appliances.

CHAPTER 2:

Automatic Smoke Detection and Exhaust Fan Control Using Arduino

2.1 Introduction

In today's digital world, electrical energy plays a vital role in powering a wide range of electronic devices such as smartphones, laptops, electric vehicles, and household appliances. The increasing dependence on electronic gadgets has created a growing demand for accessible and controlled power sources in both public and private environments.

However, in many locations such as railway stations, bus stands, commercial complexes, and educational institutions, the availability of safe and organized power supply systems is limited. Uncontrolled power outlets may lead to electricity misuse, overload conditions, and potential safety hazards.

To address these challenges, the Smart Power Nexus system is designed as an embedded-based power distribution and control platform. The system utilizes a microcontroller to regulate and control electrical power delivery through switching modules such as relays.

The Smart Power Nexus system acts as a centralized power access unit that can provide electrical power to multiple devices including mobile phones, electric vehicles, and electrical appliances. The embedded controller manages the operation of the system and ensures safe switching of AC power.

This project demonstrates the practical implementation of embedded systems in smart energy management and power automation, offering a cost-effective and scalable solution for modern infrastructure.

2.2 System Overview

The Smart Power Nexus is an embedded-based intelligent power distribution system designed to provide controlled and efficient electrical power supply to multiple connected devices.

The system mainly consists of a NodeMCU (ESP8266) microcontroller, relay switching modules, power supply circuits, and output sockets. The microcontroller acts as the central control unit that processes system operations and manages the switching of electrical power.

The relay module functions as a switching interface between the low-voltage control circuit and the high-voltage AC power supply. When the system is activated, the microcontroller sends control signals to the relay module, allowing electrical power to flow to the connected devices.

The Smart Power Nexus system performs several important functions such as:

- Providing controlled electrical power supply to connected loads.
- Managing the switching of AC power through relay modules.
- Ensuring safe operation by isolating low-voltage control circuits from high-voltage power supply.

2.3 Block Diagram

Smart Power Nexus – Embedded Based Power Distribution System

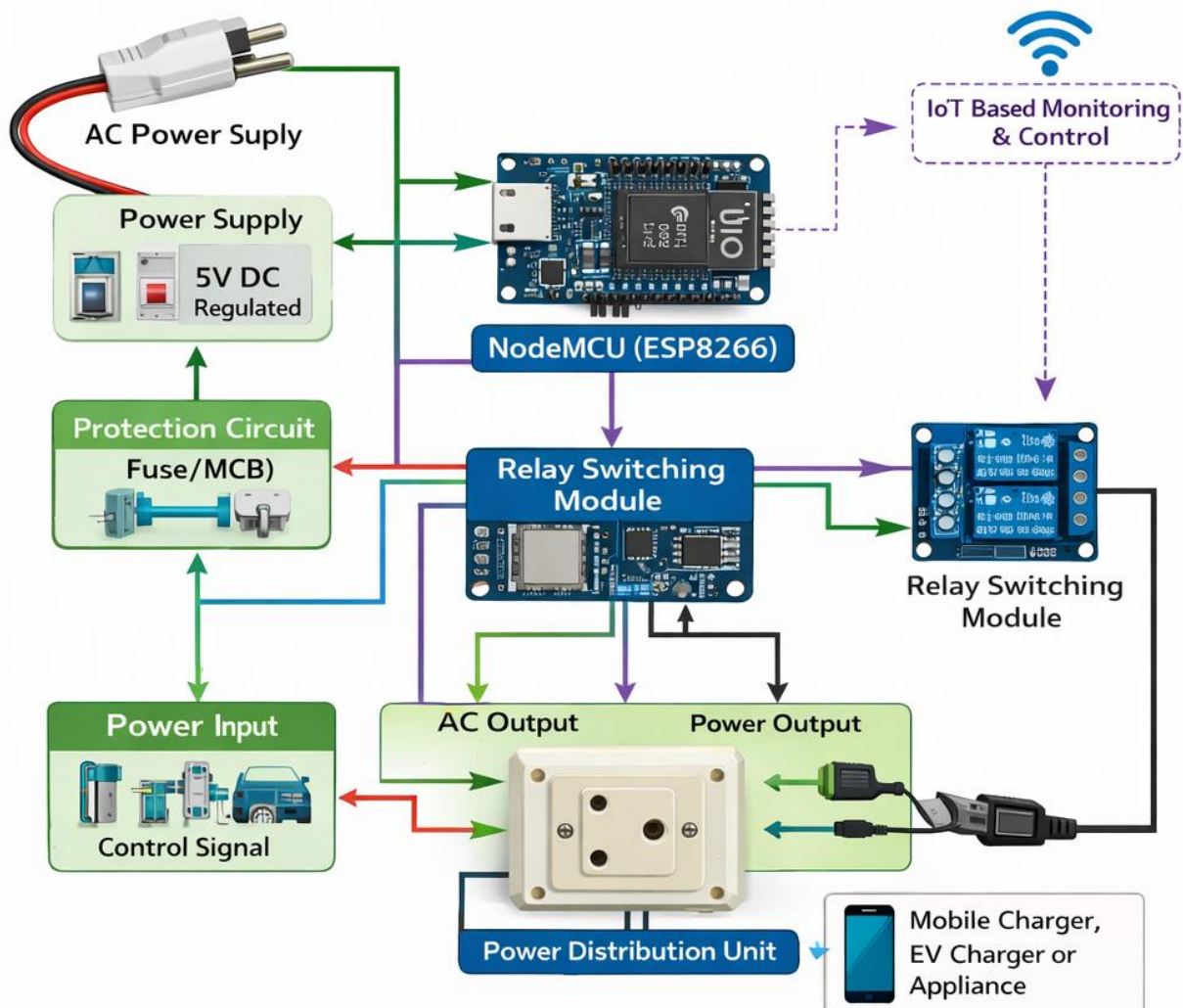


Fig 2.1 Block Diagram

Description: The block diagram illustrates Smart Power Nexus depicting the NodeMCU (ESP8266) microcontroller as the central unit controlling a relay switching module for power distribution. An AC power supply provides input, which is regulated through a protection circuit and delivered safely to a charging port.

➡ Power Input ➡ AAC Output ➡ Power Output ➡ Ground

2.4 Circuit Description

The circuit of the **Smart Power Nexus – Embedded Based Power Distribution System** is centered around the **NodeMCU (ESP8266)** microcontroller, which acts as the main control unit of the system. The NodeMCU operates at **3.3V logic level** and is responsible for controlling the switching of electrical power through a relay module.

The system uses an **IR sensor module** to detect the insertion or presence of an object (such as a coin or trigger input) in the sensing area. The IR sensor continuously monitors the input and sends a digital signal to the **NodeMCU** when an object is detected.

Table 2.1: Pin Configuration and Connected Modules

Component	Arduino UNO Pin	Function
IR Sensor Module	D5	Detects object/coin insertion
Relay Module	D1	Controls AC power switching
Power Supply	5V / VIN	Provides power to NodeMCU
AC Plug Input	AC Line	Supplies electrical power
AC Output Socket	Relay Output	Provides controlled power to devices

Working:

The **Smart Power Nexus system** operates using a combination of an embedded microcontroller, an IR sensor module, and a relay switching unit to control electrical power distribution.

Initially, the **NodeMCU (ESP8266)** microcontroller is powered through a regulated DC power supply. Once the system is powered on, the microcontroller continuously monitors the output signal from the **IR sensor module**.

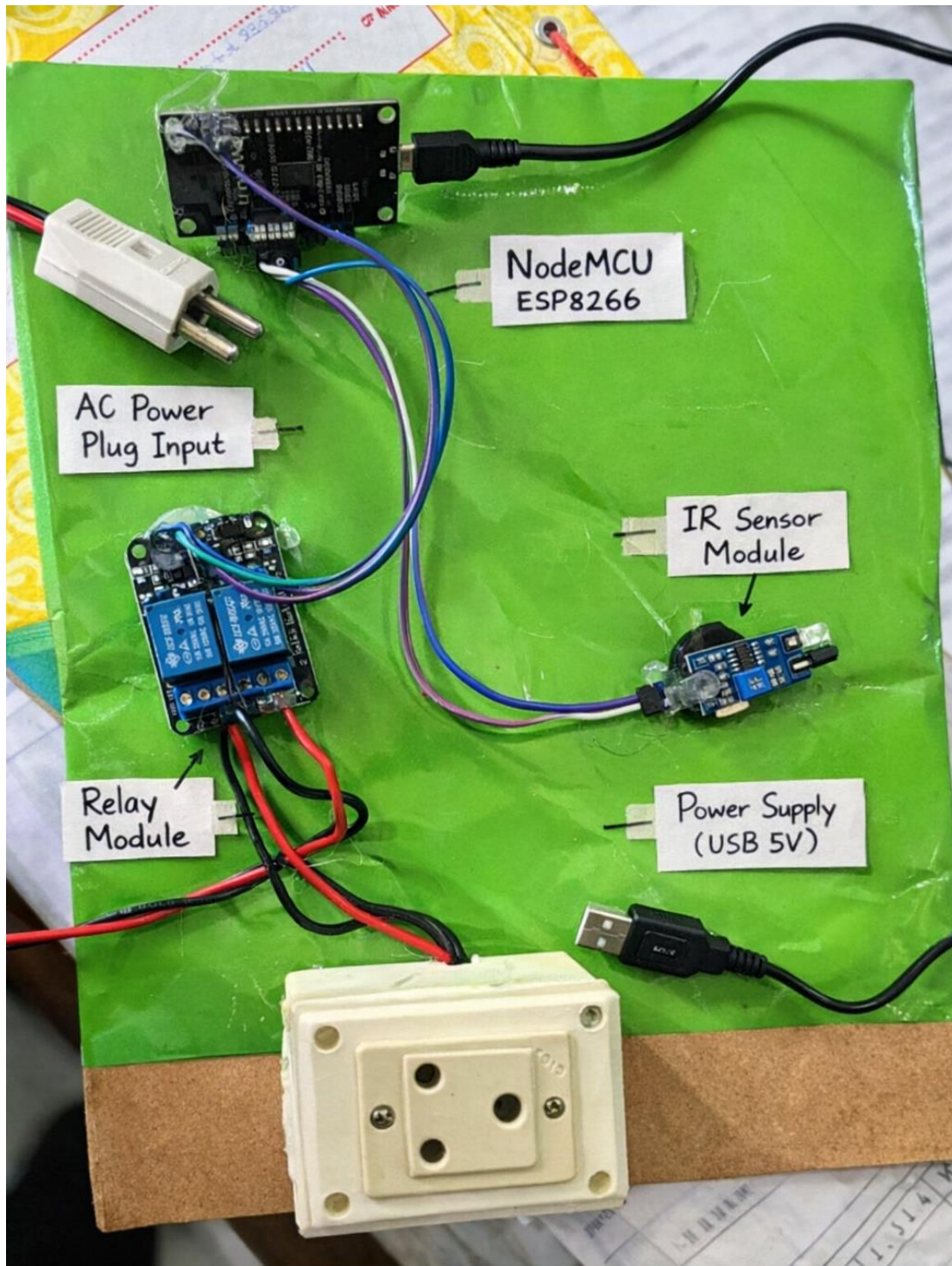


Fig 2.1 Circuit Overview

2.5 Components Explanation

2.5.1 NodeMCU (ESP8266)



Fig 2.2 NodeMCU (ESP8266)

The **NodeMCU (ESP8266)** is a WiFi-enabled microcontroller board used as the main control unit of the Smart Power Nexus system. It is based on the ESP8266 microcontroller and supports multiple digital input/output pins for interfacing with sensors and modules.

The NodeMCU processes signals received from the IR sensor and controls the relay module accordingly. It operates at a **voltage of 3.3V** and can be easily programmed using the Arduino IDE. Due to its small size, low cost, and built-in WiFi capability, NodeMCU is widely used in embedded systems and IoT applications.

In this project, the **NodeMCU** acts as the brain of the system by controlling the power switching operation and ensuring automated power distribution.

2.5.2 IR Sensor Module

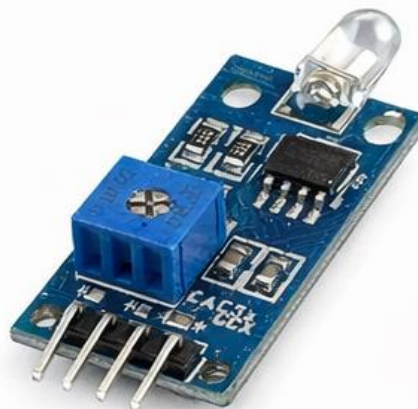


Fig 2.4 IR Sensor Module

The **IR (Infrared) Sensor Module** is used for object detection in the system. It works by transmitting infrared light and detecting the reflected signal from nearby objects.

When an object such as a coin passes through the sensor area, the reflected infrared light is detected by the receiver, and a digital signal is sent to the **NodeMCU**. The microcontroller then processes this signal and triggers the relay module.

The IR sensor operates at **3.3V–5V** and provides both analog and digital outputs. It is commonly used in automation systems for object detection and proximity sensing.

In this project, the IR sensor detects the input trigger and activates the smart power distribution process.

2.5.3 Relay module

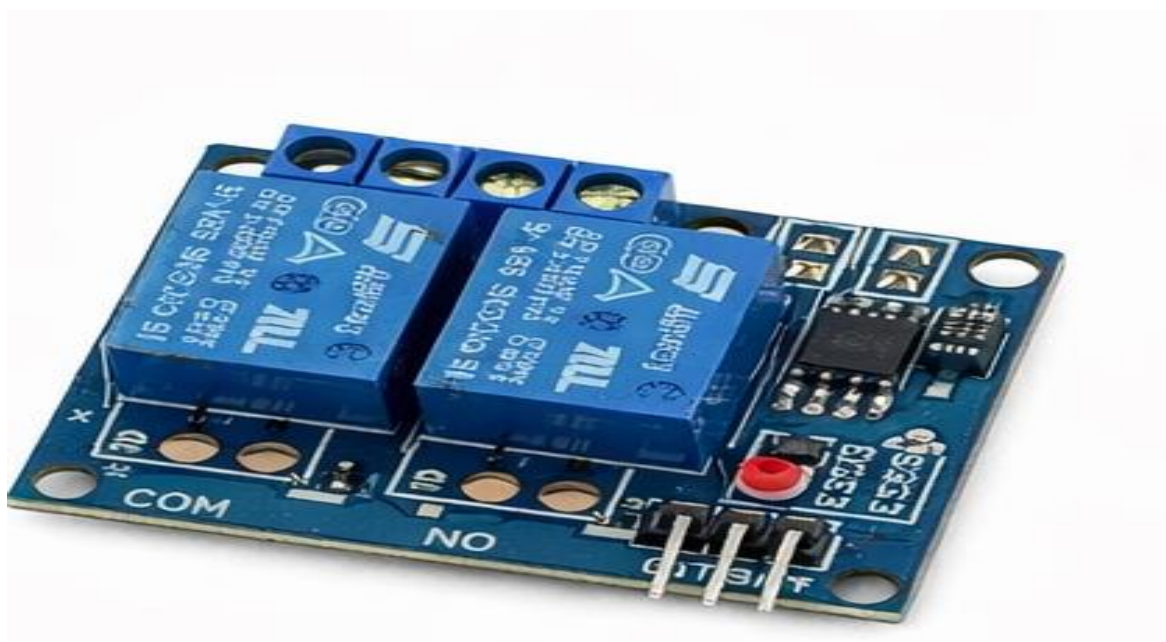


Fig 2.5 Relay module

A **relay module** is an electrically operated switch that allows a low-voltage microcontroller circuit to control high-voltage electrical devices.

The relay module used in this project is a **5V** dual-channel relay capable of switching AC loads safely. When the NodeMCU sends a control signal to the relay module, the relay activates and connects the AC power supply to the output socket.

This ensures electrical isolation between the low-voltage control circuit and the high-voltage AC supply, improving safety and reliability.

In this system, the relay module controls the power supply to the AC socket where external electrical devices are connected.

2.5.4 AC Socket Output



Fig 2.6 AC Socket Output

The **AC socket** acts as the output interface of the Smart Power Nexus system. Electrical devices such as mobile chargers or appliances can be connected to this socket.

When the relay module is activated, the AC power supply flows through the relay contacts and reaches the output socket. When the relay is turned off, the power supply to the socket is disconnected.

This mechanism ensures controlled and safe power delivery to connected devices.

2.5.5 Power Supply



Fig 2.7 Power Supply

The **power supply** unit provides the required electrical power to operate the NodeMCU and other electronic components in the system.

In this project, a USB **5V** power source is used to power the NodeMCU microcontroller. The regulated power ensures stable operation of the embedded system.

The AC supply connected through the plug is used only for powering external electrical devices through the relay-controlled socket.

2.6 Software Implementation

The **Smart Power Nexus system** is programmed using the Arduino IDE, which provides a simple environment for writing, compiling, and uploading code to the NodeMCU (ESP8266) microcontroller. The NodeMCU reads the input signal from **the IR sensor module** and processes it according to the predefined program logic. When the IR sensor detects an object or trigger input, the microcontroller activates the **relay module**, which connects the AC power supply to the output socket. This allows electrical devices such as mobile chargers or appliances to receive power.

The program continuously monitors the sensor status and controls the relay module accordingly. If the sensor input changes or the programmed condition ends, the NodeMCU deactivates the relay module, automatically disconnecting the power supply. This ensures controlled and safe power distribution.

The Arduino IDE is used to write the embedded **C/C++ code**, compile it, and upload it to the NodeMCU through a **USB cable** connection. The software implementation ensures reliable communication between the sensor, controller, and switching module.

Key Libraries Used:

- **<ESP8266WiFi.h>** — Used for enabling WiFi features of the NodeMCU (optional for IoT upgrades).
- **<Arduino.h>** — Provides the basic functions required for Arduino-based programming.
- **<Wire.h>** — Used for communication with additional modules if required in future upgrades.

2.7 Flowchart

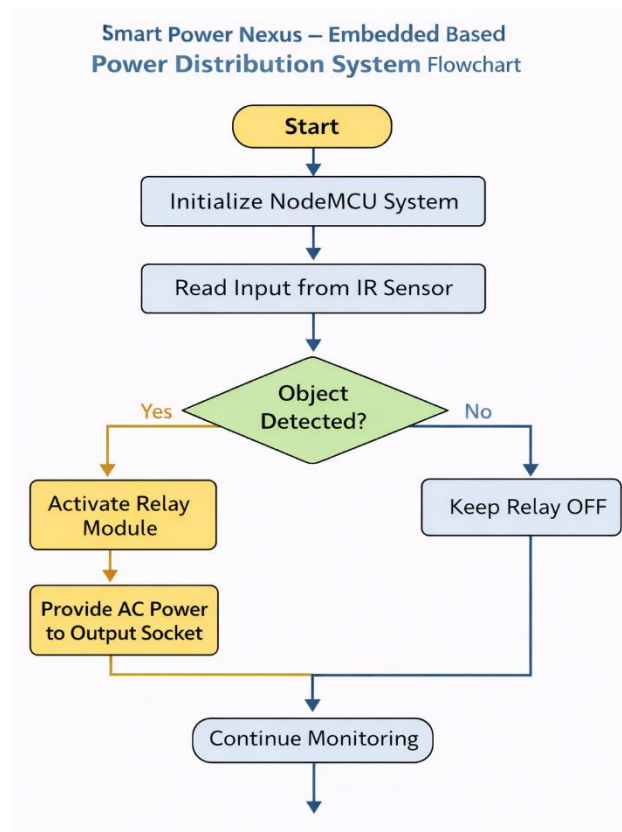


Fig 2.8 Flowchart of the system

2.8 Working Principle.

1. When the power supply is provided, the **NodeMCU (ESP8266)** microcontroller initializes the system and prepares all connected modules for operation.
2. The **IR sensor module** continuously monitors the sensing area to detect the presence of an object or trigger input.
3. When an object is detected, the sensor sends a signal to the **NodeMCU microcontroller**.
4. The NodeMCU processes the signal and activates the **relay module**, which connects the AC power supply to the output socket.
5. The connected electrical device such as a **mobile charger or appliance** receives power through the AC socket.
6. The system continuously monitors the sensor status and controls the relay accordingly to ensure safe and controlled power distribution.

CHAPTER 3:

RESULTS AND DISCUSSION

3.1 Introduction

After the successful design and development of the Smart Power Nexus – Embedded Based Power Distribution System, testing was carried out to evaluate the performance, reliability, and efficiency of the system. The objective of testing was to ensure that the system operates correctly according to the programmed logic and provides controlled power supply to connected devices.

This chapter presents the results obtained from testing individual modules as well as the complete integrated system. The analysis focuses on the performance of the IR sensor module, NodeMCU microcontroller, and relay switching module in controlling the electrical power supply.

The evaluation mainly focuses on:

- The response time of the IR sensor module in detecting an object.
- The ability of the NodeMCU microcontroller to process input signals correctly.
- The proper operation of the relay module in switching AC power supply.
- The overall system performance in providing controlled power output.
- The reliability and stability of the system during continuous operation.

3.2 Experimental Setup

The prototype of the Smart Power Nexus system was developed using a NodeMCU (ESP8266) microcontroller board, which acts as the central control unit of the system. The system was powered using a USB 5V power supply, which provides the required operating voltage for the microcontroller and sensor modules.

An IR sensor module was used to detect the presence of an object or trigger input. The sensor continuously monitored the sensing area and transmitted signals to the NodeMCU when an object was detected.

A relay module was connected to the NodeMCU to control the switching of the AC power supply. The relay module acts as an interface between the low-voltage control circuit and the high-voltage AC power source.

The AC plug input provided the electrical power source, while the AC output socket served as the output interface where electrical devices such as mobile chargers or appliances could be connected.

The entire system was assembled on a prototype board and tested under different conditions to evaluate the performance and reliability of the smart power distribution mechanism. The testing confirmed that the system successfully controlled the power supply based on the sensor input and operated as expected.

3.3 .1Testing of Individual Modules

3.3.1 IR Sensor Detection Module

The IR sensor module was tested to verify its ability to detect objects placed in front of the sensor. During testing, different objects were moved across the sensing area to evaluate the response of the sensor.

The sensor successfully detected the presence of objects and generated a digital signal which was transmitted to the NodeMCU microcontroller. The response time of the sensor was found to be quick and reliable.

Table 3.1: IR Sensor Detection Test Results

Test Condition	Object Presence	Sensor Output	system Status
No Object	Absent	LOW	System Idle
Object Detected	Present	HIGH	Trigger Signal
Continuous Detection	Present	HIGH	Active State

Observation:

The IR sensor module responded quickly when an object entered the sensing area. The sensor output changed from LOW to HIGH, allowing the NodeMCU to process the signal and activate the relay module.

3.3.2 Relay Control Module

The **relay module** was tested to verify its ability to control the AC power supply based on signals received from the NodeMCU.

When the IR sensor detected an object, the NodeMCU sent a control signal to the relay module, activating the relay switch. This allowed the AC power supply to flow to the output socket.

Table 3.2: Relay Module Test Results

Sensor Condition	Relay State	Power Output	System Status
No Object	OFF	No Power	Idle
Object detected	ON	Power Supplied	Active
Object Removed	OFF	Power Disconnected	Standby

Observation:

The relay module successfully switched the AC power supply ON and OFF according to the control signals from the NodeMCU, ensuring safe and controlled power distribution.

NodeMCU Control Module

The NodeMCU (ESP8266) microcontroller was tested to ensure proper communication between the IR sensor and the relay module.

The microcontroller successfully received input signals from the sensor and processed them according to the programmed logic. It then generated appropriate output signals to control the relay module.

Table 3.4: NodeMCU Module Test Results

Input Signal	Processing	Relay Output	System Status
LOW	No Action	OFF	Idle
HIGH	Signal Processed	ON	Power Activated
Continuous Signal	Active Processing	ON	System Running

Observation:

The NodeMCU performed reliably during testing and successfully coordinated the operation of the sensor and relay modules.

3.4 Integrated System Testing

After verifying the independent operation of each module, all components were integrated to form the complete Smart Power Nexus – Embedded Based Power Distribution System. The system was tested under different practical conditions to evaluate its functionality, reliability, and response.

Test Scenario 1 – Normal Safe Condition

- No object detected by the IR sensor
- Relay module remains in OFF state
- AC output socket does not receive power

Result:

The system remained in standby mode. The relay module stayed OFF and no electrical power was supplied to the output socket.

Test Scenario 2 – Object Detection

- An object was placed in front of the IR sensor
- The IR sensor detected the object and sent a signal to the NodeMCU
- NodeMCU processed the signal and activated the relay module

Result:

The relay module switched ON and allowed AC power to flow to the output socket. The connected electrical device received power successfully.

Test Scenario 3 – Object Removed

- The object was removed from the IR sensor detection area
- The IR sensor output returned to its normal state

Result:

The NodeMCU deactivated the relay module and the AC power supply to the output socket was disconnected automatically.

Observation:

The integrated Smart Power Nexus system operated successfully under all test conditions. The IR sensor responded quickly to object detection, and the NodeMCU correctly controlled the relay module based on the sensor input.

The system demonstrated reliable performance, safe power switching, and effective control of electrical power distribution.

3.5 System Output

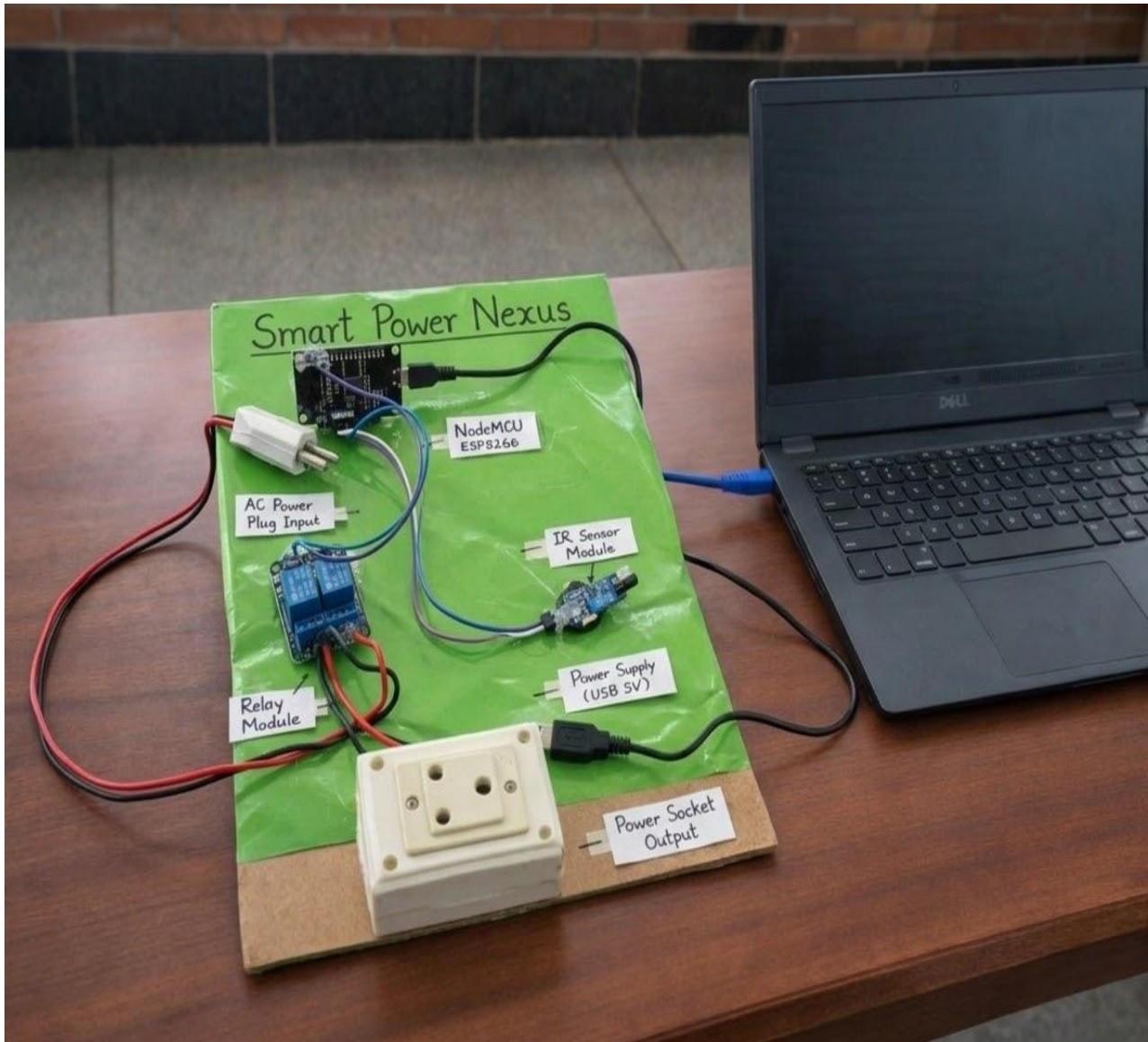


Fig 3.1 System Output

3.6 Performance Analysis

The performance of the Smart Power Nexus system was evaluated based on several operational parameters such as response time, reliability, and system stability.

- **Response Time:** The IR sensor detected objects quickly and transmitted signals to the NodeMCU almost instantly.
- **Accuracy:** The sensor detection and relay switching worked accurately during repeated testing.
- **Reliability:** The system operated continuously without malfunction during multiple testing cycles.
- **Power Consumption:** The NodeMCU and sensor modules consumed minimal power, making the system energy efficient.

Table 3.4: System Performance Parameters

Parameter	Measured Value	Remarks
Object Detection Delay	< 1 second	Fast response
Relay Activation Time	< 1 second	Smooth switching
Power Control Accuracy	High	Reliable operation
System Stability	Continuous operation	Stable performance

3.7 Discussion

From the experimental results and system testing, the following observations were made:

1. The IR sensor successfully detected objects placed in the sensing area and transmitted signals to the NodeMCU microcontroller.
2. The NodeMCU processed the sensor signals efficiently and controlled the relay module according to the programmed logic.
3. The relay module reliably switched the AC power supply ON and OFF, enabling controlled power distribution.
4. The integrated system demonstrated stable operation during repeated testing conditions.
5. The Smart Power Nexus system provides a simple, cost-effective, and efficient solution for automated power distribution using embedded system technology.

3.8 Limitations

Although the Smart Power Nexus system performed effectively during testing, a few practical limitations were observed:

- The IR sensor may be affected by strong external light sources.
- The system is currently implemented as a prototype model and requires proper enclosure for real-world deployment.
- Additional protection circuits may be required for high-power electrical appliances.
- The system currently provides basic power control and may require advanced monitoring features for large-scale applications.

CHAPTER 4:

CONCLUSION AND FUTURE SCOPE

4.1 Conclusion

The Smart Power Nexus – Embedded Based Power Distribution System was successfully designed and implemented to provide controlled electrical power supply using embedded system technology.

The system integrates a NodeMCU (ESP8266) microcontroller, an IR sensor module, and a relay switching module to automate the process of power distribution. The IR sensor detects the presence of an object or trigger input and sends a signal to the NodeMCU microcontroller. Based on this input, the microcontroller activates or deactivates the relay module to control the AC power supply connected to the output socket.

Testing results showed that the system operates reliably and responds quickly to sensor inputs. The relay module effectively controlled the power supply, ensuring safe and efficient power distribution. The developed prototype demonstrates that embedded systems can be effectively used to implement smart power management solutions for public charging stations, smart infrastructure, and automated power control applications.

4.2 Future Scope

The Smart Power Nexus system can be further enhanced with additional technologies to improve functionality and usability. The following improvements can be implemented in future versions:

1. IoT Integration:

The system can be connected to the internet using the built-in WiFi capability of the NodeMCU for remote monitoring and control.

2. Mobile Application Control:

A smartphone application can be developed to monitor system status and control power distribution remotely.

3. Digital Payment Integration:

Integration of **UPI, QR code, or digital payment systems** can allow users to access power through secure payment methods.

4. Energy Monitoring System:

Current and voltage sensors can be integrated to measure and monitor power consumption of connected devices.

5. Multiple Charging Ports:

The system can be expanded to support multiple output sockets to charge several devices simultaneously.

6. Smart Timer Control:

A timer-based control system can be added to automatically disconnect power after a specified time.

7. Overload Protection System:

Additional protection circuits such as overload protection and circuit breakers can be implemented to ensure electrical safety.

8. Smart Display Interface:

An LCD or OLED display can be integrated to show system status, power usage, and operational information.

9. Remote Monitoring Dashboard:

A web-based dashboard can be developed to monitor system activity and performance in real time.

10. Voice Control Integration:

The system can be integrated with smart assistants such as **Google Assistant or Alexa** for voice-based control.

11. Smart Energy Management:

Advanced algorithms can be implemented to optimize energy distribution and improve efficiency.

12. Industrial-Grade Enclosure Design:

The prototype can be redesigned with a compact and protective enclosure for real-world deployment in public and commercial environments.

ANNEXURE

CODE

```
#define IR_SENSOR D2
#define RELAY D1

unsigned long chargingTime = 60000;
bool charging = false;

void setup()
{
  pinMode(IR_SENSOR, INPUT);
  pinMode(RELAY, OUTPUT);
  digitalWrite(RELAY, HIGH);
  Serial.begin(9600);
}

void loop()
{
  int sensorState = digitalRead(IR_SENSOR);

  if(sensorState == LOW && charging == false)
  {
    Serial.println("Coin Detected");
    charging = true;

    digitalWrite(RELAY, LOW);
    Serial.println("Charging ON");

    delay(chargingTime);

    digitalWrite(RELAY, HIGH);
    Serial.println("Charging OFF");

    charging = false;
    delay(2000);
  }
}
```

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